

SWENG3043

Operating Systems

Software Engineering Department
AASTU

Chapter Nine

Storage Management

Chapter Objectives

- To describe the physical structure of secondary storage devices and its effects on the uses of the devices.
- To explain the performance characteristics of mass-storage devices.
- To evaluate disk scheduling algorithms.

Topics Included

- Disk Hardware
- Disk scheduling
- Disk management

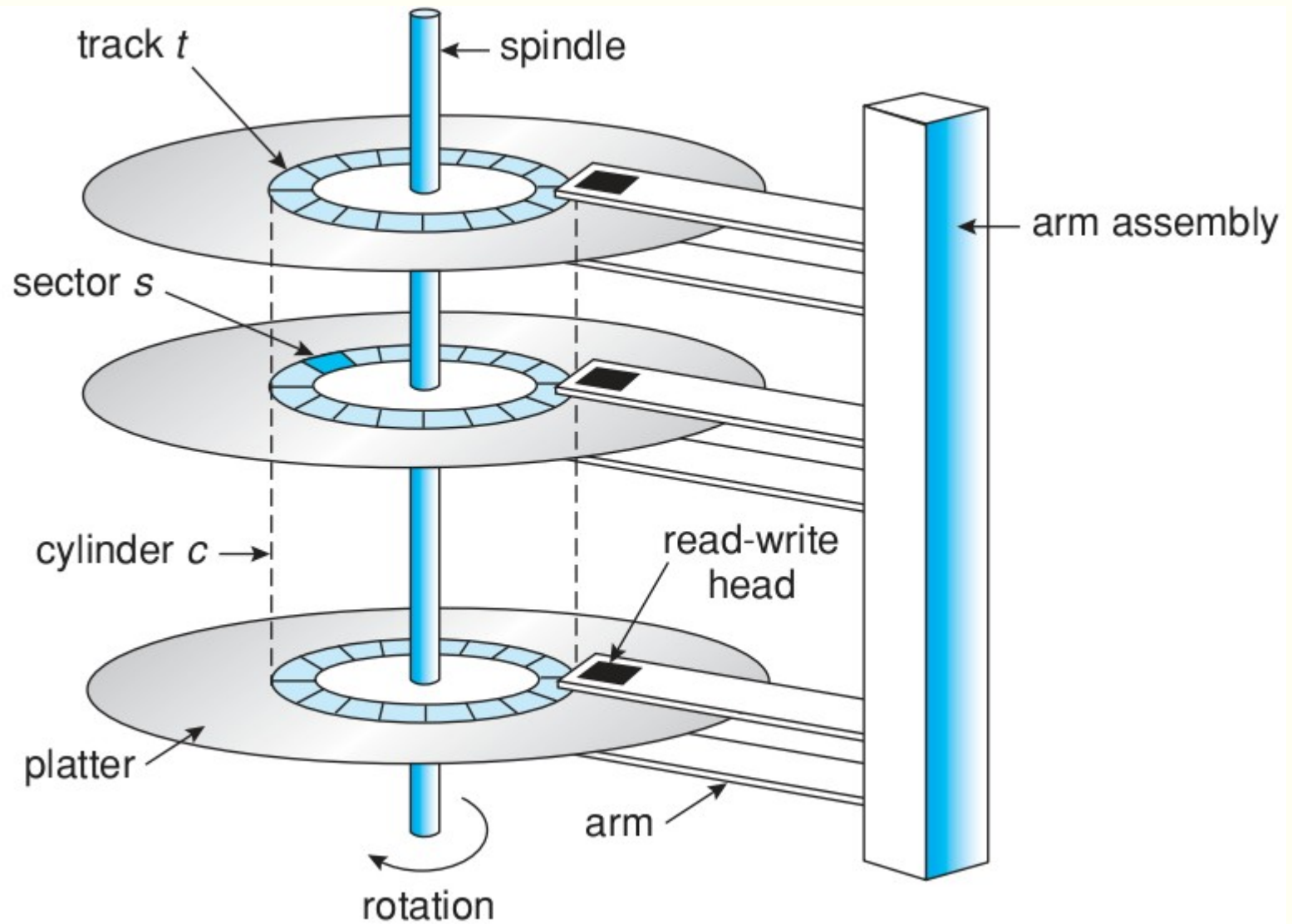
Disk Hardware

- All modern computers except embedded ones have disk drives.
- All real disks are organized into cylinders, each one containing many **tracks**.
- Each of the tracks then will be divided into **sectors**
- The set of tracks that are at one arm position makes up a **cylinder**.
- Common platters range 3.5", 2.5", and 1.8"

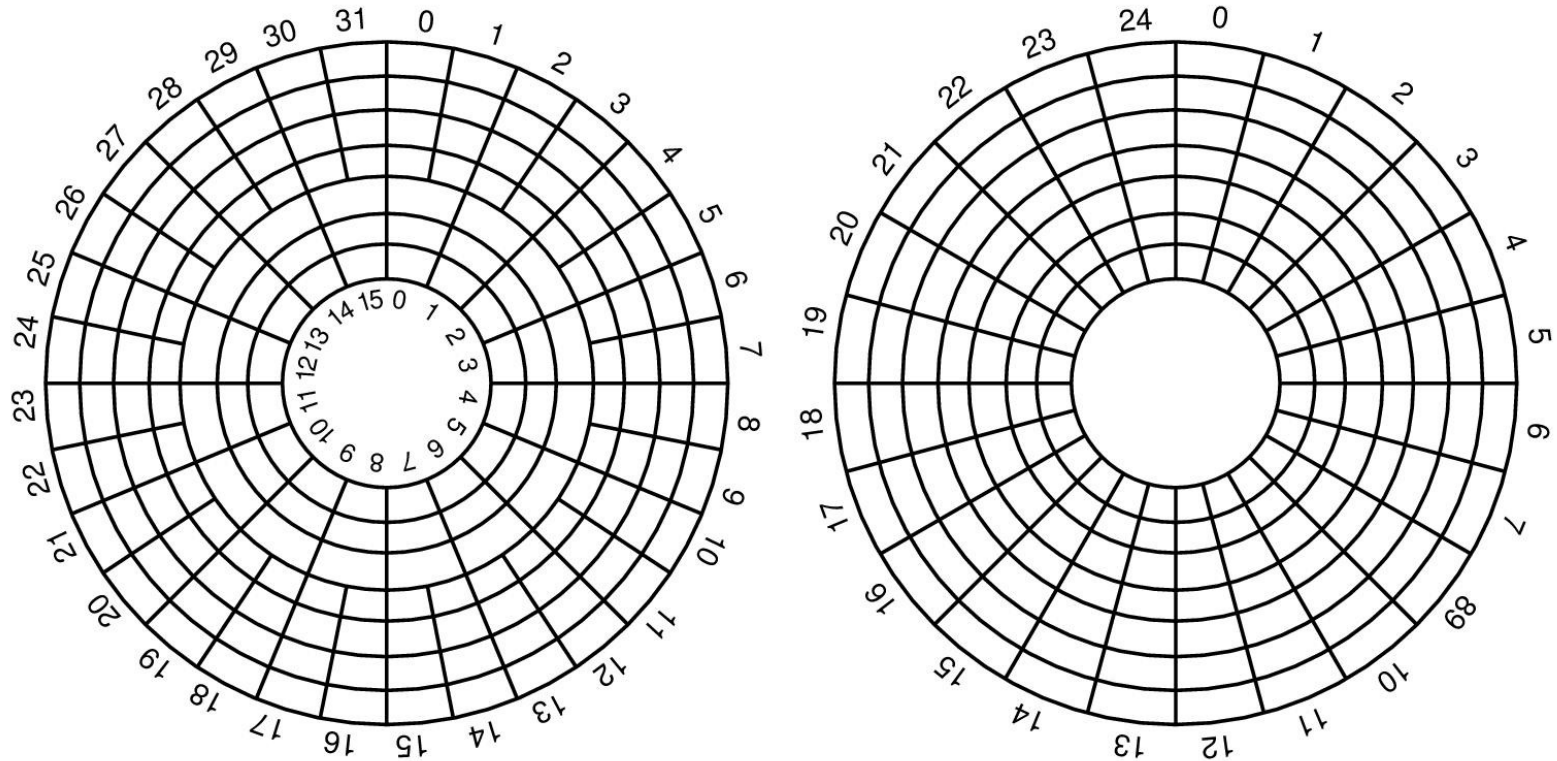
al number of sectors

The data density as closer to the center (hub) is high

The speed increases as the read/write moves to the outer tracks



Disk Hardware(cont'd)



- Physical geometry of a disk with two zones
- A possible virtual geometry for this disk

Disk Access Time

- HDDs rotate at 60 to 250 times per second
- The time required to read or write a disk block is determined by three factors:
 - **Transfer rate** is rate at which data flow between drive and computer
 - **Positioning time** is time to move disk arm to desired cylinder (**seek time**) and time for desired sector to rotate under the disk head (**rotational latency**)
- For most disks, the seek time dominates the other two times, so reducing the mean seek time can improve system performance substantially.
- **Head crash** results from disk head making contact with the disk surface
 - That's bad

A 3.5" HDD with Cover Removed.



The First Commercial Disk Drive



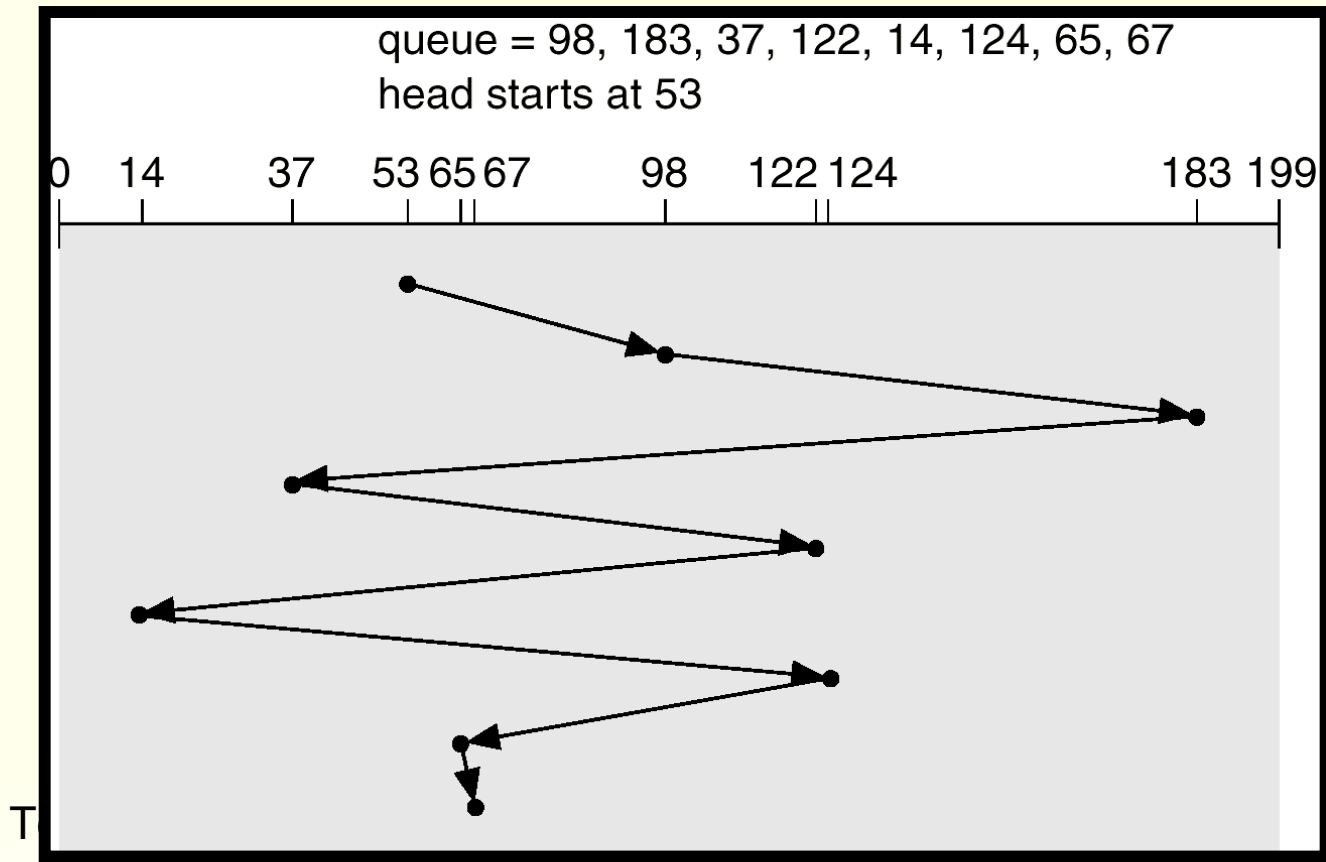
- 1956
- IBM RAMDAC computer included the IBM Model 350 disk storage system
- 5 million (7 bit) characters
- 50 x 24" platters
- Access time = < 1 second

Disk Arm Scheduling Algorithm

- The OS maintains queue of requests for each I/O operation
- It uses disk scheduling algorithm
- **First Come First Served (FCFS)** : Accept a request one at a time and carries them out in that order
 - **E.g 1.** Consider a disk queue with requests for I/O to blocks on cylinders: 1, 36, 16, 34, 9, 12
Head starts at 11
Service order: 1,36,16,34,9,12
Arm motion required: 10, 35, 20, 18, 25, 3,
Total= 111 cylinders.
- The simplest and the fairest of all, but it doesn't improve performance

FCFS(cont'd)

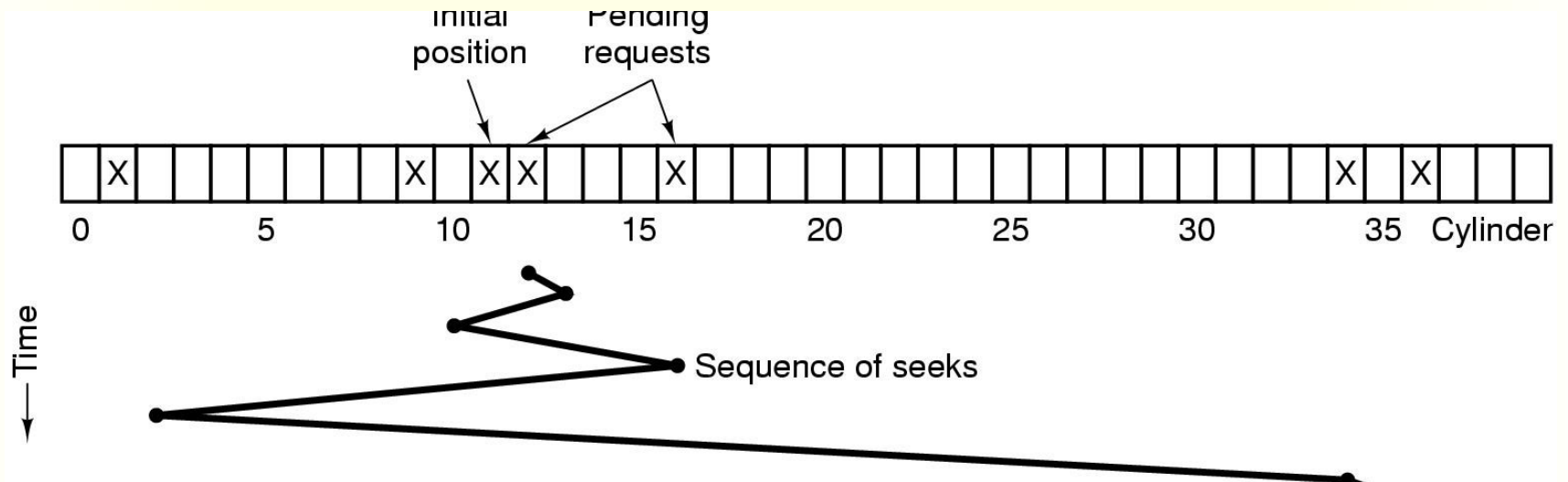
- **E.g 2.** Consider a disk queue with requests for I/O to blocks on cylinders:
98, 183, 37, 122, 14, 124, 65, 67



Shortest Seek First(SSF)

- It handles the closest (the least disk arm movement) request next, to minimize seek time.
 - **E.g. 1** Consider a disk queue with requests for I/O to blocks on cylinders: 1, 36, 16, 34, 9, 12
 - Head starts at 11
 - Service order: 12,9,16, 1, 34, 36
 - Arm motion required: 1, 3, 7, 15, 33, 2
 - Total= 61
- Advantage: Performance (efficiency), provides better performance
- Disadvantage: Possibility of starvation (it lacks fairness)

SSF(cont'd)

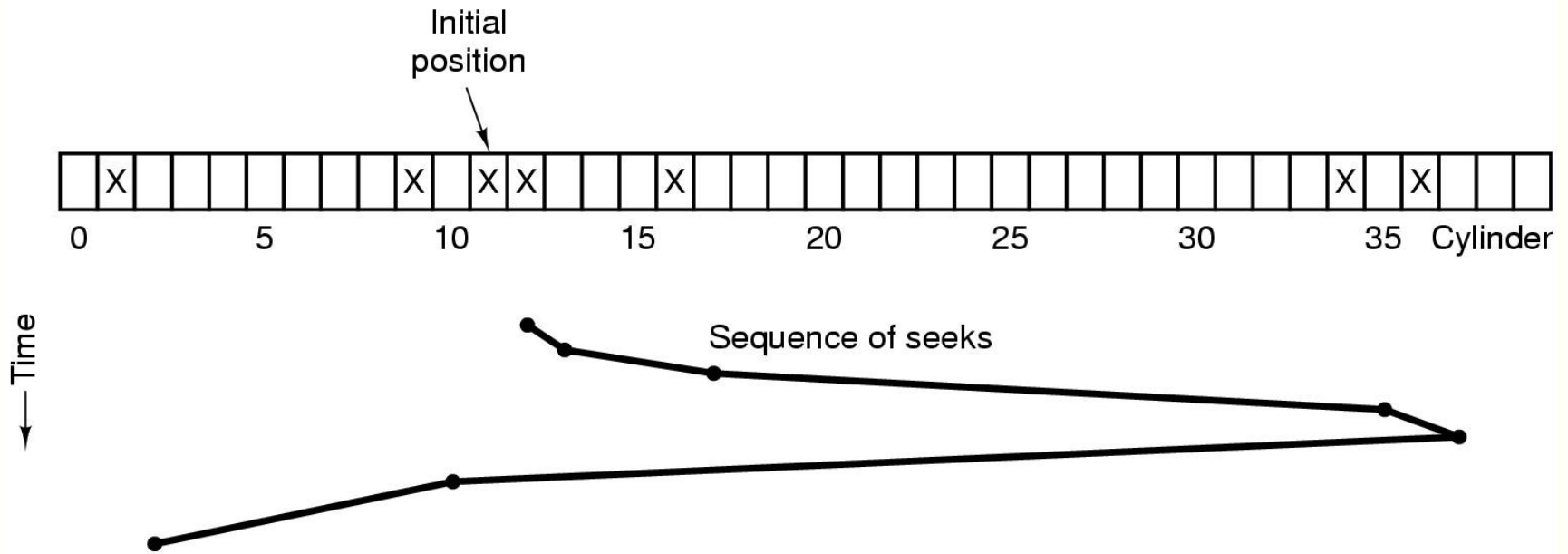


Shortest Seek First (SSF) disk scheduling algorithm

SCAN (Elevator) Algorithm

- **SCAN (Elevator) Algorithm** : The disk arm keeps moving in the same direction until there are no more outstanding requests in that direction, then it switches direction
- Direction bit 1= up, 0=down
 - E.g. Track initial position: 11
 - Track request: 1, 36, 16, 34, 9, 12
 - Direction bit: 1
 - Service order: 12, 16, 34, 36, 9, 1
 - Disk arm motion: 1, 4, 18, 2, 27, 8
 - Total= 60 cylinders

SCAN Algorithm(cont'd)

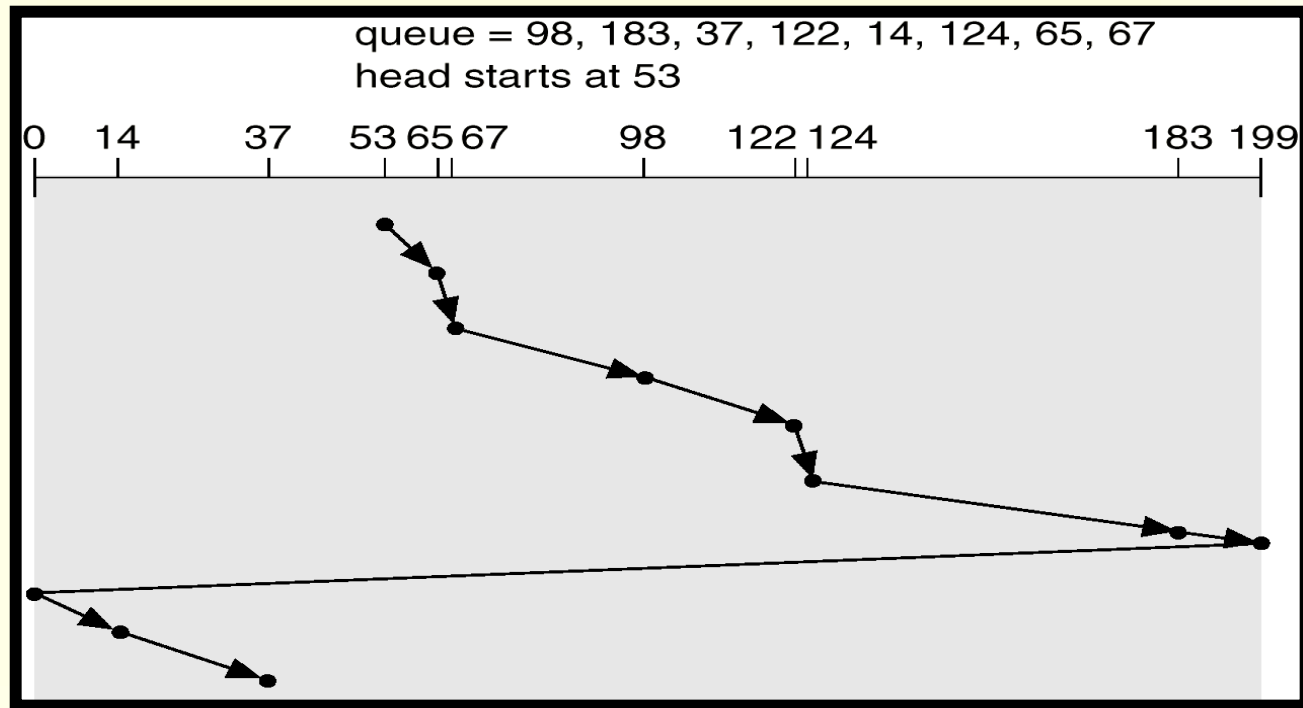


The elevator algorithm for scheduling disk requests

C-SCAN (Modified Elevator) Algorithm

- Provides a more uniform wait time than SCAN
- The head moves from one end of the disk to the other, servicing requests as it goes
 - When it reaches the other end, however, it immediately returns to the beginning of the disk, without servicing any requests on the return trip
- Treats the cylinders as a circular list that wraps around from the last cylinder to the first one
- It reduces the maximum delay experienced by new request

C-SCAN Algorithm(cont'd)



Selecting Disk- Scheduling Algorithm

- SSF is common and has a natural appeal because it increases performance over FCFS
- SCAN and C-SCAN perform better for systems that place a heavy load on the disk.
 - Less starvation
- Performance depends on the number and types of requests.
- Requests for disk service can be influenced by the file-allocation method.
- The disk-scheduling algorithm should be written as a separate module of the operating system, allowing it to be replaced with a different algorithm if necessary.

Disk Management

■ Disk Formatting

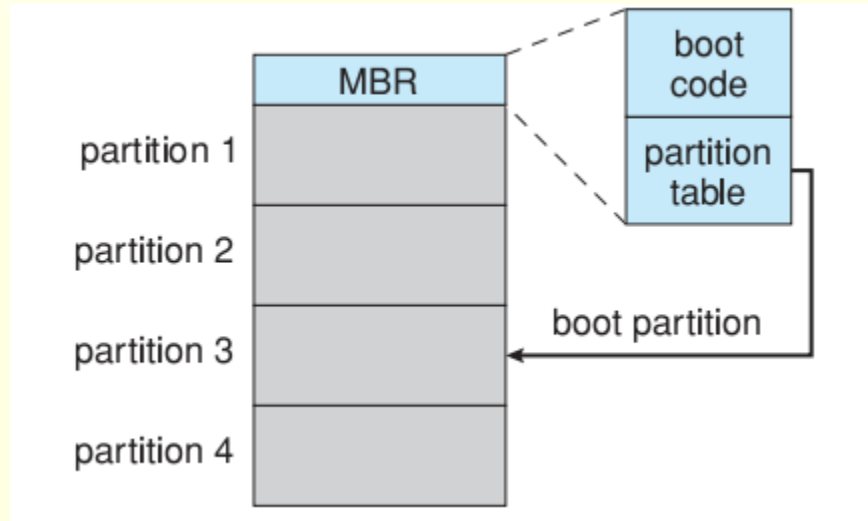
- A new magnetic disk is a blank slate. Before a disk can store data, it must be divided into sectors that the disk controller can read and write.
- Low-level formatting, or physical formatting — Dividing a disk into sectors that the disk controller can read and write.
- To use a disk to hold files, the operating system still needs to record its own data structures on the disk.
 - Partition the disk into one or more groups of cylinders.
 - Logical formatting or “making a file system”.
 - operating system stores the initial file-system data structures onto the disk

■ Boot block initializes system.

- The bootstrap is stored in ROM.
 - initializes all aspects of the system, from CPU registers to device controllers and the contents of main memory, and then starts the operating system.
- A disk that has a boot partition is called a boot disk or system disk.

■ Methods such as sector sparing used to handle **bad blocks**.

Booting from disk in Windows



- boot partition— contains the operating system and device drivers
- Master boot record, or MBR – holds Windows boot code